SkillBridge: An AI-Based Career Guidance and Recruitment Platform

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Abstract— Online recruitment platforms have significantly improved access to employment opportunities; however, many existing systems provide limited support for personalized career guidance, skill gap analysis, mentorship, and intelligent recruitment services within a single platform. This paper presents SkillBridge, an AI powered web-based career guidance and recruitment platform that integrates job seekers, employers, and career mentors into a unified ecosystem. The proposed system provides AI based job recommendations, explainable recommendation results, skill gap analysis, resume evaluation, mentorship support, interview management, and credential verification using SHA 256 fingerprints. The system is designed using software engineering principles and modeled through Use Case, Class, and Deployment Diagrams to ensure a scalable and maintainable architecture. A prototype of SkillBridge was implemented using PHP and MySQL, with AI assisted modules supporting intelligent career recommendations, resume analysis, interactive dashboards, and recruitment management. The platform also provides administrative analytics to assist employers and administrators in monitoring recruitment activities and system performance. Experimental outputs demonstrate that the platform successfully integrates multiple career development services, provides transparent recommendation results, and improves recruitment efficiency through a centralized web application. Compared with existing recruitment platforms, SkillBridge offers a more comprehensive solution by combining intelligent recommendation, career guidance, resume evaluation, credential verification, and employer recruitment management within a single system. The proposed platform demonstrates the potential of Artificial Intelligence to improve employability, simplify recruitment processes, and support data driven career development.

Index Terms— Artificial Intelligence, Career Guidance, Recruitment System, Recommender System, Explainable AI, Skill Gap Analysis, Resume Evaluation.

I. Introduction

The rapid growth of Artificial Intelligence (AI) and recommender systems has transformed modern recruitment and career guidance. Traditional job portals primarily connect employers with job seekers but often lack personalized career support, skill-gap analysis, and mentorship services. Recent research has shown that AI-driven recommendation systems can significantly improve job matching by analyzing users' skills, qualifications, and labor market requirements, leading to more accurate and personalized career recommendations [1], [2].

Despite these advancements, current recruitment platforms still face several limitations. Many systems provide general job recommendations without identifying users' missing skills or offering personalized career guidance. In addition, mentorship, resume evaluation, and employer recruitment are usually provided as separate services rather than within one integrated platform. Research also indicates that traditional recommendation methods struggle to adapt quickly to changing user preferences and newly available job opportunities, reducing recommendation effectiveness [2], [3].

The main contributions of this work are summarized as follows:

- Development of SkillBridge, an AI-powered career guidance and recruitment platform that integrates Job Seekers, Career Mentors, and Employers/HR within a unified ecosystem.
- Integration of intelligent features including AI-based job recommendations, skill-gap analysis, resume evaluation, mentorship support, and recruitment management to improve employability and hiring efficiency.
- Design of a modular software architecture using Use Case, Class, and Deployment Diagrams, providing a scalable foundation for future development.

II. Related Work

Artificial Intelligence (AI), recommender systems, and intelligent recruitment platforms have become important research areas in recent years due to their ability to improve

career guidance and recruitment efficiency. For this study, the literature review focused on AI-based job recommendation systems, personalized career guidance, and intelligent recruitment platforms. The selected papers were chosen because they are recent, peer-reviewed publications from reputable journals, including IEEE Access and Elsevier, and are closely related to the objectives of the proposed SkillBridge system.

Siswipraptini et al. [1] proposed a personalized career-path recommendation model for Information Technology students using AI-based techniques to recommend suitable career paths according to students' academic backgrounds and skills. The model demonstrated effective career recommendations but focused only on career planning without integrating employer recruitment or mentorship services.

Tavakoli et al. [2] developed an AI-based open recommender system for personalized labor market-driven education, enabling learners to receive recommendations based on labor market demands. Although the proposed system improved educational guidance, it did not provide an integrated recruitment platform connecting job seekers, employers, and mentors.

Kwieciński et al. [3] introduced a comparison between real-time and batch job recommendation approaches to improve recommendation quality and system responsiveness. Their study achieved better recommendation performance but concentrated mainly on recommendation algorithms without incorporating additional career development features such as mentorship, resume evaluation, or skill-gap analysis.

Although these studies have contributed significantly to AI-powered recruitment and recommendation systems, several research gaps remain. Most existing solutions focus on a single aspect of recruitment, such as career recommendation, educational guidance, or job matching. They rarely integrate multiple services, including AI-powered job recommendations, skill-gap analysis, mentorship, employer recruitment, and resume evaluation, within a single platform. To address these limitations, this study proposes SkillBridge, an integrated AI-powered career guidance and recruitment platform that combines these features to improve employability, recruitment efficiency, and user experience.

III. Problem Statement

The rapid growth of online recruitment platforms has made job searching more accessible; however, many existing systems still fail to provide personalized career guidance and comprehensive recruitment support. Most recruitment platforms primarily focus on posting job vacancies and receiving applications, offering limited assistance in identifying users' skill gaps, recommending suitable career paths, or providing mentorship. As a result, job seekers often struggle to find opportunities that match their qualifications, while employers spend considerable time screening large numbers of applications.

Another major challenge is the lack of integration among different career development services. Career guidance, resume evaluation, mentorship, and recruitment management are often provided through separate platforms, requiring users to switch between multiple systems. This fragmented approach reduces efficiency, increases user effort, and limits the overall recruitment experience. Furthermore, many traditional recruitment systems face challenges in accurately matching job seekers with suitable opportunities due to the complexity of representing users' skills, experiences, and job requirements [4]**.**

To address these challenges, this study proposes SkillBridge, an AI-powered career guidance and recruitment platform that integrates Job Seekers, Employers/HR, and Career Mentors into a unified system. The platform provides AI-based job recommendations, skill-gap analysis, resume evaluation, mentorship support, and recruitment management within a single web application. By integrating these services, SkillBridge aims to improve recruitment efficiency, enhance employability, and provide a more personalized and intelligent career development experience.

IV. Proposed Solution

A. System Overview

The proposed system, SkillBridge, is an AI-powered web-based career guidance and recruitment platform designed to bridge the gap between job seekers, employers, and career mentors. The platform integrates recruitment, career development, and mentorship services into a single application, enabling users to access personalized career support throughout the employment process. By utilizing Artificial Intelligence (AI) and modern recommendation techniques, the system provides AI-powered job recommendations with Explainable AI (XAI), allowing users to understand why specific jobs are recommended. The platform also performs skill-gap analysis, resume evaluation, and recruitment assistance to improve both employability and hiring efficiency. Recent research has shown that integrating large language models with recommender systems can better capture user preferences and content semantics, leading to more effective recommendations [5].

The system consists of three primary actors: Job Seeker**,** Employer/HR, and Career Mentor. Job seekers can create profiles, upload resumes, search and apply for jobs, receive AI-

powered recommendations, and communicate with career mentors. Employers can post job vacancies, review applications, shortlist candidates, and schedule interviews. Career mentors can monitor job seekers' progress, provide career guidance, review resumes, and offer personalized feedback. The interaction among these actors enables a collaborative recruitment environment that supports both career development and organizational hiring needs.

The proposed system adopts a modular architecture to ensure scalability, maintainability, and future extensibility. AI services are integrated with the application through dedicated modules, allowing intelligent recommendations and analytical features to operate efficiently while maintaining a clear separation between application logic, database management, and AI components.

B. Use Case Diagram

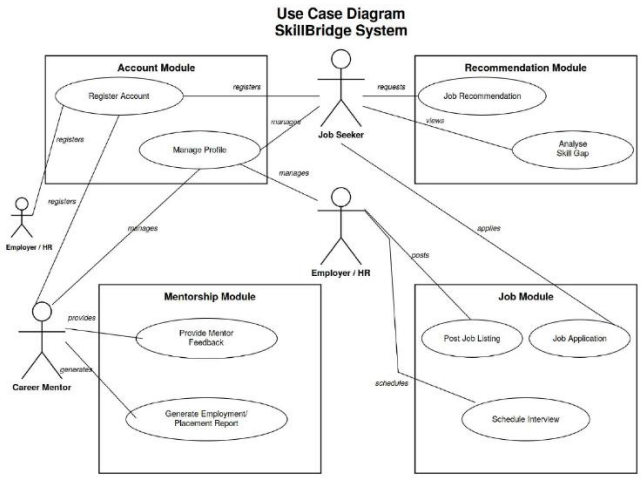


Fig. 1. Use Case Diagram of the SkillBridge System.

The Use Case Diagram illustrates the interactions between the three primary actors—Job Seeker**,** Employer/HR**,** and Career Mentor—and the SkillBridge system. It presents the major functionalities available to each actor, including user authentication, profile management, job posting, job applications, AI-powered job recommendations, mentorship, interview scheduling, and recruitment management. This diagram provides a high-level overview of the system's functional requirements and user interactions.

C. Class Diagram

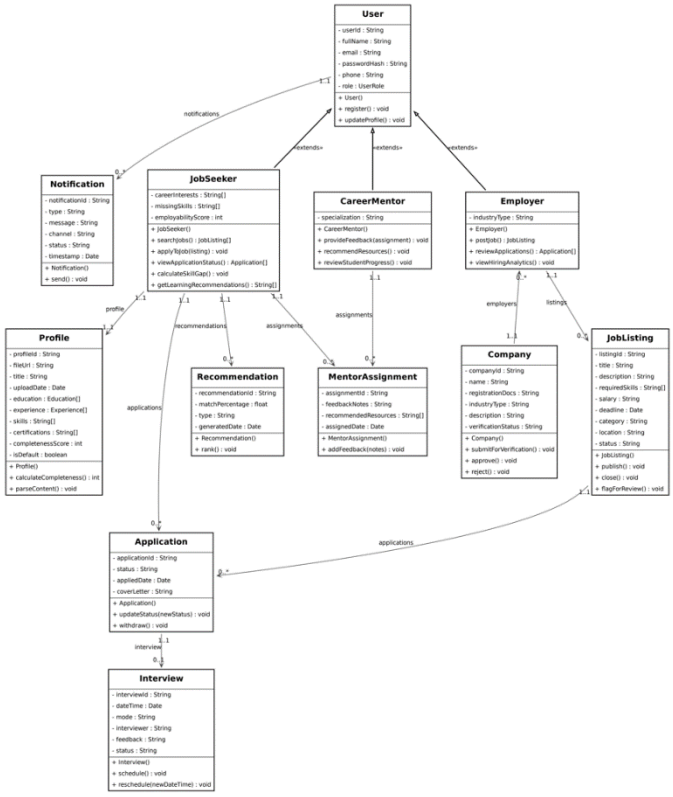


Fig. 2. Class Diagram of the SkillBridge System.

The Class Diagram represents the static structure of the SkillBridge system by illustrating the main classes, their attributes, operations, and relationships. The diagram defines the interactions among entities such as Job Seeker, Employer, Career Mentor, Job Listing, Application, Recommendation, Company, and Profile. These relationships establish the foundation of the system's object-oriented design and ensure efficient data management and software maintainability.

D. Deployment Diagram

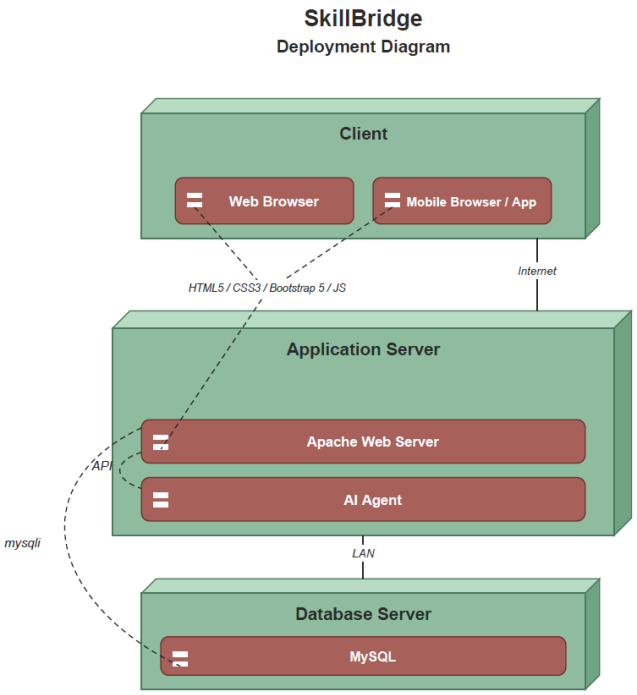


Fig. 3. Deployment Diagram of the SkillBridge System.

The Deployment Diagram illustrates the physical architecture of the SkillBridge system. It consists of client devices connected through the Internet to the application server, which hosts the web application and communicates with the database server for data storage and retrieval. An AI agent is integrated with the application server through APIs to provide intelligent services such as job recommendation, skill gap analysis, and resume evaluation. This architecture ensures scalability, modularity, and efficient communication between system components.

V. Result Analysis

This section presents the outputs generated by the proposed SkillBridge system. The screenshots demonstrate the major functionalities, including AI-powered recommendations, resume analysis, credential verification, analytics, and career guidance services.

A. Student Dashboard

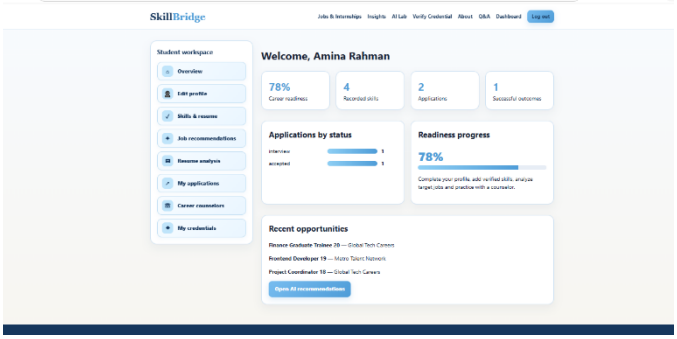


Fig. 4. Student dashboard.

Figure 4 shows the student dashboard after login. It displays career readiness, recorded skills, submitted applications, successful outcomes, and recent job opportunities. The dashboard provides students with an overview of their career progress and quick access to important platform features.

B. AI-powered Job Recommendation

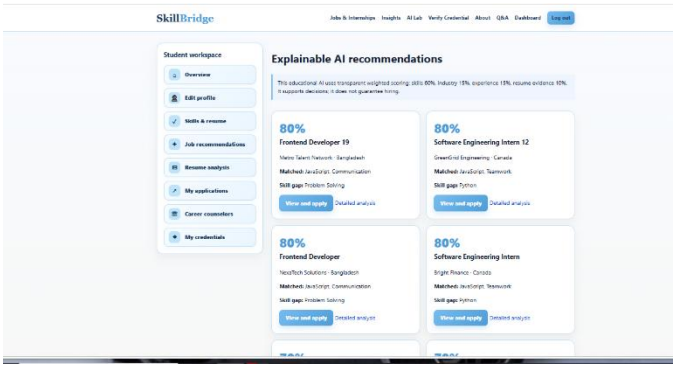


Fig. 5. Explainable AI job recommendation.

Figure 5 presents the AI-powered job recommendation module. The system recommends suitable jobs based on user skills and experience while providing explainable AI (XAI) information, including matched skills and identified skill gaps, enabling users to understand why each recommendation is generated.

C. AI-assisted Resume Analysis

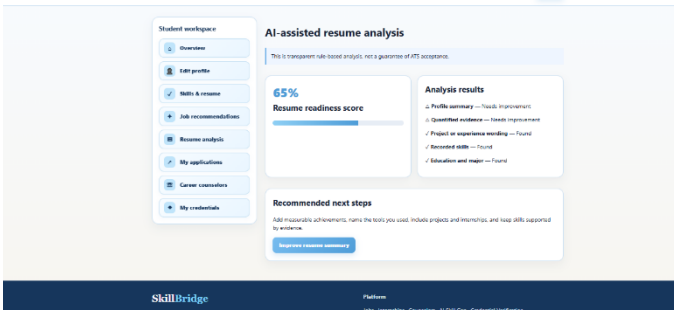


Fig. 6. AI-assisted resume analysis.

Figure 6 illustrates the resume analysis module. The system evaluates resume readiness, highlights missing information,

and provides recommendations to improve the quality of the user's resume before applying for jobs.

D. AI Classifier Laboratory

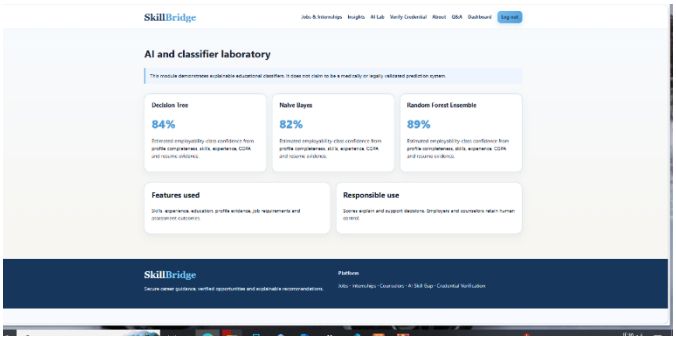


Fig. 7. AI classifier laboratory.

Figure 7 compares the performance of three machine learning classifiers: Decision Tree, Naïve Bayes, and Random Forest. The module estimates employability confidence using profile completeness, skills, education, and resume evidence, while supporting transparent AI-based decision making.

E. Credential Verification

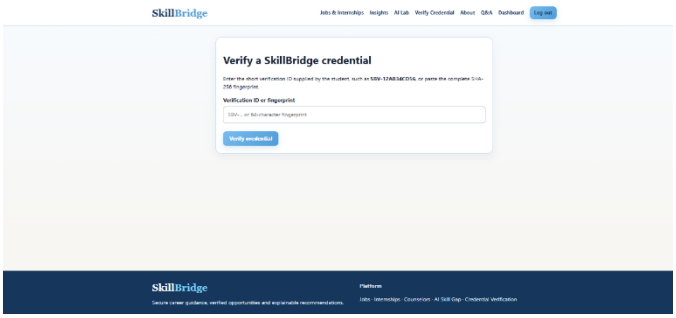


Fig. 8. Credential verification using SHA-256.

Figure 8 demonstrates the credential verification module. Users can verify certificates by entering a verification ID or SHA-256 fingerprint, ensuring the authenticity and integrity of uploaded credentials.

F. Administration Dashboard

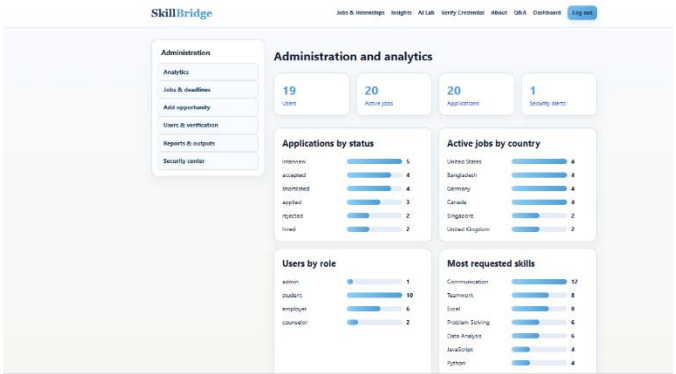


Fig. 9. Administration dashboard.

Figure 9 presents the administration dashboard, which provides real-time analytics on users, job postings, applications, security alerts, and requested skills. These analytical reports help administrators monitor system activities and support data-driven decision making.

VI. Comparative Analysis

Table I compares the proposed SkillBridge system with several widely used recruitment platforms. While existing platforms provide job searching and recruitment services, they offer limited support for integrated career development features. The proposed SkillBridge system combines AI-powered job recommendations, skill-gap analysis, mentorship, resume evaluation, and interview management within a single platform, providing a more comprehensive solution for both job seekers and employers.

Table I. Comparison of SkillBridge with Existing Recruitment Platforms

Feature	LinkedIn	Indeed	Glassdoor	SkillBridge (Proposed)
Job Search	✓	✓	✓	✓
Job Application	✓	✓	✓	✓
AI Job Recommendation	Limited	Limited	✗	✓
Skill Gap Analysis	✗	✗	✗	✓
Career Mentorship	✗	✗	✗	✓
Resume Evaluation	Limited	Limited	✗	✓
Interview Scheduling	Limited	✓	✗	✓
Employer Recruitment Management	✓	✓	Limited	✓
Integrated Career Development	✗	✗	✗	✓

The comparison demonstrates that existing recruitment platforms mainly focus on job searching and application management. In contrast, SkillBridge integrates recruitment, AI-powered recommendations, skill-gap analysis, mentorship, resume evaluation, and employer management into a single platform. This integrated approach improves the overall recruitment process while providing personalized career guidance for job seekers.

VII. Conclusion

This paper presented SkillBridge, an AI-powered career guidance and recruitment platform that integrates job seekers, employers, and career mentors within a single web-based system. The proposed platform combines AI-based job recommendations, skill-gap analysis, mentorship support, resume evaluation, and recruitment management to improve both employability and hiring efficiency. The system was

designed using software engineering principles and modeled through Use Case, Class, and Deployment Diagrams to ensure a scalable and maintainable architecture.

In the future, SkillBridge can be enhanced by incorporating advanced AI models for more accurate recommendations, real-time labor market analytics, chatbot-based career assistance, and integration with professional networking platforms. These improvements would further enhance user experience and increase the effectiveness of intelligent recruitment and career development services. Overall, the proposed SkillBridge platform demonstrates the potential of integrating AI-driven career guidance and recruitment services within a unified web-based system.

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